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EDITORIAL

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A post-fire management strategy for the EU

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Abstract

Wildfires across the European Union are increasing in frequency, extent, and severity, driven by climate change, prolonged droughts, and shifting land-uses. Although substantial progress has been made in prevention and suppression, post-fire management remains the least developed component in the fire management cycle, despite its critical role in mitigating cascading impacts such as soil erosion, nutrient losses, invasive species expansion, and the degradation of ecosystem services. This work synthesizes current knowledge, operational practices, and policy frameworks related to post-fire management in the EU, drawing on scientific literature and expert experience from Greece, Portugal, and Spain. We review available tools, such as stabilization measures, restoration techniques, and monitoring, and identify a persistent implementation gap between scientific knowledge and on-the-ground practice. Building on this analysis, we propose a strategic pathway toward a EU-level post-fire management framework grounded in risk-based prioritization, coordinated monitoring, and stronger integration between science and policy. The roadmap highlights common assessment procedures, multidisciplinary operational teams, long-term funding structures, and knowledge exchange across Member States, while retaining the flexibility required to reflect regional ecological and institutional diversity. We emphasize that Europe already possesses the scientific and technical foundations necessary to implement effective post-fire management. The key challenge now lies in institutionalizing these tools within coherent governance structures. Advancing a coordinated, evidence-based framework would not only enhance recovery after wildfires but also strengthen landscape resilience, climate adaptation, and the long-term sustainability of European ecosystems.

Resumen

Los incendios de vegetación en Unión Europea están aumentando en frecuencia, extensión, y severidad, debidos fundamentalmente al Cambio Climático, sequías prolongadas, y cambios en el uso de la tierra. Aunque se han hecho progresos sustanciales en prevención y supresión, el manejo post-incendio permanece como el componente menos desarrollado del ciclo de la gestión del fuego, a pesar de su rol crítico en mitigar la cascada de impactos que estos conllevan, como la erosión del suelo, la pérdida de nutrientes, la expansión de de especies invasoras, y la degradación de los servicios ecosistémicos. Este trabajo sintetiza el conocimiento actual, las prácticas operacionales y los marcos conceptuales de las políticas relacionadas con el manejo post-incendio en la Unión Europea, basados en la literatura científica y experiencias de expertos de Grecia, Portugal, y España. Revisamos las herramientas disponibles como medidas de estabilización, técnicas de restauración y monitoreo, e identificamos el persistente vacío de implementación entre el conocimiento científico y las prácticas operativas a campo. En este análisis, proponemos un camino

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estratégico hacia un marco conceptual de manejo post-incendio a nivel Europeo basado en la priorización del riesgo, el monitoreo coordinado, y una fuerte integración entre ciencia y política. La hoja de ruta subraya la determinación de procedimientos comunes, la formación de grupos operacionales multidisciplinarios, la creación de estructuras de financiación a largo plazo, y el intercambio de conocimiento entre los Estados miembros, manteniendo la flexibilidad necesaria para reflejar la diversidad ecológica e institucional de cada estado. Enfatizamos que Europa posee las capacidades científicas y técnicas necesarias para implementar una gestión efectiva post-incendio. El desafío clave descansa ahora en institucionalizar estas herramientas dentro de una estructura de gobernanza coherente. El avance sobre un marco conceptual coordinado, basado en evidencia científica, podría mejorar la recuperación post-incendio, y también fortalecería la resiliencia de los ecosistemas, la adaptación al cambio climático y sostenibilidad a largo plazo de los ecosistemas europeos.

Introduction

Wildfires in the European Union (EU) are becoming increasingly severe and widespread, driven by climate change, prolonged heatwaves and droughts, and ongoing land-use changes. In 2025 alone, wildfires burned 1,031,405 ha across the EU, nearly three times the 2006–2024 annual average of 354,842 ha (San-Miguel-Ayanz et al. 2025), causing profound environmental and societal impacts, such as an alarming biodiversity loss in protected areas (Thom and Seidl 2016; Roces-Díaz et al. 2022; Driscoll et al. 2024). Economically, the agriculture and forestry sectors have suffered substantial losses, while tourism has been disrupted across fire-prone regions such as southern France, Portugal, Spain, and Greece (Neger et al. 2024; Abrhám et al. 2025).

To effectively respond to the scale and severity of wildfires across the EU, it is necessary to adopt an integrated fire management framework that systematically connects prevention, preparedness, suppression, and post-fire recovery (Held et al. 2026). This approach places post-fire mitigation, ecological restoration, and long-term risk reduction on equal footing with preventive and suppressive measures within EU wildfire policy.

Wildfires have lasting impacts both within and beyond the burned areas. The elimination of the topsoil protective layer through the combustion of vegetation and litter, combined with the fire-induced changes in soil properties, such as altered soil structure and the development of soil water repellency (Robichaud 2025), can lead to increased surface runoff and accelerated erosion (Vieira et al. 2023). These fire-induced changes can further impact soil health by amplifying nutrient and carbon losses (Girona-García et al. 2024), compromising key ecosystem services, including land productivity, flood regulation, and clean water provision (Vieira et al. 2023; Girona-García et al. 2024).

However, once a fire has occurred and its damage has been assessed, the work is far from complete. Post-fire management is essential not only for effective ecosystem recovery, but also for improving landscape resilience and reducing the likelihood of future wildfires (Moreira et al. 2012a; Barros et al. 2018; Tubbesing et al. 2019; Lucas-Borja et al. 2021). Yet, evidence from previous studies and EU-level assessments shows that post-fire management actions in the EU are often limited in spatial extent and inconsistently applied (Vieira et al. 2023). For example, emergency stabilization techniques (mulching, construction of log erosion barriers or check dams) are often limited to easily accessible locations or to protecting priority infrastructure, leaving large portions of burned catchments untreated. This may amplify post-fire disturbances, as untreated areas remain significant sources of runoff and sediment (Moreira et al. 2012b; Vieira et al. 2023). Moreover, current post-fire management practices frequently focus on short-term objectives and small-scale applications, aimed at preventing immediate damage, such as erosion or flooding (Menéndez-Miguélez et al. 2025). Implementation decisions are often made under significant time pressure (ECA 2025), which further constrains the adoption of systematic and ecologically grounded approaches.

In the EU, long-term post-fire management initiatives are constrained by fragmented regulatory frameworks and predominantly short-term policy objectives, lacking an effective ecological planning (Vallejo et al. 2012; Moreira et al. 2020). Moreover, administrative procedures for securing post-fire management funding, particularly at local and regional levels, are highly complex (ECA 2025) and often become available too late to achieve effective outcomes (Ribeiro et al. 2020). Consequently, actions are often guided more by technical experience than by scientific evidence, resulting in a trial-and-error approach in which tradition and urgency outweigh systematic planning and evidence-based practices (Guimarães et al. 2024). Moreover, the lack of assessments on the effectiveness of implemented measures prevents errors from being corrected in future interventions.

Post-fire management is further constrained if conducted on private land by the heterogeneous land uses, vegetation, landowner interests, and applicable legal frameworks. Unclear rules and legislative hurdles often

shift costs to landowners, reducing the motivation and effectiveness of interventions (Palaiologou et al. 2018; Fernandes and Simões 2024). This fragmented governance can limit coordinated, landscape-scale responses at a national level, and further increase complexity on post-fire management decisions at EU scale (ECA 2025). Consequently, burned areas may experience neglect, mismanagement, or delayed mitigation efforts, ultimately compromising ecosystem resilience and amplifying exposure to secondary hazards.

In various regions (North America, South America) formalized post-fire response plans have been widely institutionalized (Robichaud and Ashmun 2012; Souza-Alonso et al. 2022). Conversely, across most of Europe and in many other world regions, post-fire management remains fragmented, characterized by a heterogeneous mix of tools and regionally tailored approaches rather than a consistent, unified standard (Vega et al. 2013). However, established operational post-fire emergency response systems follow a common framework, which, despite local characteristics, can be effectively transferred to other governmental contexts. Examples of those are present in global initiatives like the FAO Fire Management Hub (FAO 2023), identifying common framework design principles (risk assessment, ecological restoration, monitoring, and stakeholder involvement) that could be translated into structured frameworks at the EU scale. Here we advocate for a timely, coordinated, and evidence-based framework for post-fire management, aligned with ambitions in the EU Forest Strategy (EC 2021a), EU Biodiversity strategy (EC 2020), and EU Soil strategy (EC 2021b) but also novel legislative tools such as the Nature Restoration Regulation (EC 2024) and the Soil Monitoring Law (EC 2025).

The objective of this work is to delineate a pathway for the implementation of a post-fire management framework in the EU. To this end, we first identify existing tools and available knowledge, as well as the current obstacles in the implementation of post-fire management actions, drawing on experience from countries where our research is based, namely Greece, Portugal, and Spain, that when combined, also represent a substantial share of EU burned area annually (63.8%, average from 2006 to 2024 (San-Miguel-Ayanz et al. 2025)). We then outline a strategic way forward grounded in risk assessment and monitoring, with the aim of bridging the gap between scientific knowledge, policy making and operational application.

Tools and knowledge

Until recently, European and national public authorities responsible for land and fire management often pointed to the lack of tools for an informed land management decision-making process (Lopes 2025). As a result, the development of technical capacity and guidance within EU-funded Research and Innovation (R&I) initiatives has increased substantially (Vega et al. 2013; Karetsos et al. 2014; Valor et al. 2023), closely tied to spatial planning and remote sensing technologies for wildfire and post-fire risk assessment and monitoring (San-Miguel-Ayanz et al. 2025). Despite the strong emphasis on fire prevention and monitoring in recent years (ECA 2025), the post-fire phase has received significantly less attention. Although multi-year monitoring is crucial for understanding the effectiveness of post-fire management, EU programs and assessments typically pay little attention to burned-area recovery beyond the first post-fire year (Vieira et al. 2023).

Post-fire decision-making by land managers and competent authorities (Fig. 1) typically begins with risk assessment, which demands field data collection to evaluate fire-induced damage and potential exposure. Based on this assessment, a range of management actions may be implemented, from hillslope erosion control to the protection of downstream values at risk. Management responses are further shaped by local legislation, which may mandate specific procedures for handling fire-affected timber, managing invasive species, or conducting reforestation. Throughout this process, systematic monitoring tools are essential for evaluating progress and effectiveness and, when initial measures prove inadequate, for informing the adaptation of strategies to support long-term recovery. (Fig. 1).

Models for post-fire risks associated with soil erosion, flooding and debris flows, have been applied across local, national, EU, and global scales (Parente et al. 2022; Vieira et al. 2023, 2026b). Following risk identification, management efforts should be directed toward high-risk areas to implement appropriate stabilization, rehabilitation, and restoration strategies (Robichaud and Ashmun 2012). A diverse range of mitigation measures is required to reduce post-fire impacts, ideally being implemented immediately after fire. On fire-affected hillslopes, stabilization treatments that increase protective ground cover, such as the application of wood residues or straw mulch, are among the most cost-effective methods for mitigating post-fire soil erosion (Girona-García et al. 2021, 2023). Within drainage channels, interventions may include debris removal, structural reinforcement, and check dams to control runoff and sediment transport (Aristeidis and Vasiliki 2015; Papaioannou et al. 2023). Transportation infrastructure, including roads and trails, also requires targeted protection through actions such as resizing or removing culverts and installing barriers to prevent washouts and blockages (Napper 2006).

Post-fire salvage logging is commonly carried out to support economic recovery, safety, and fuel management

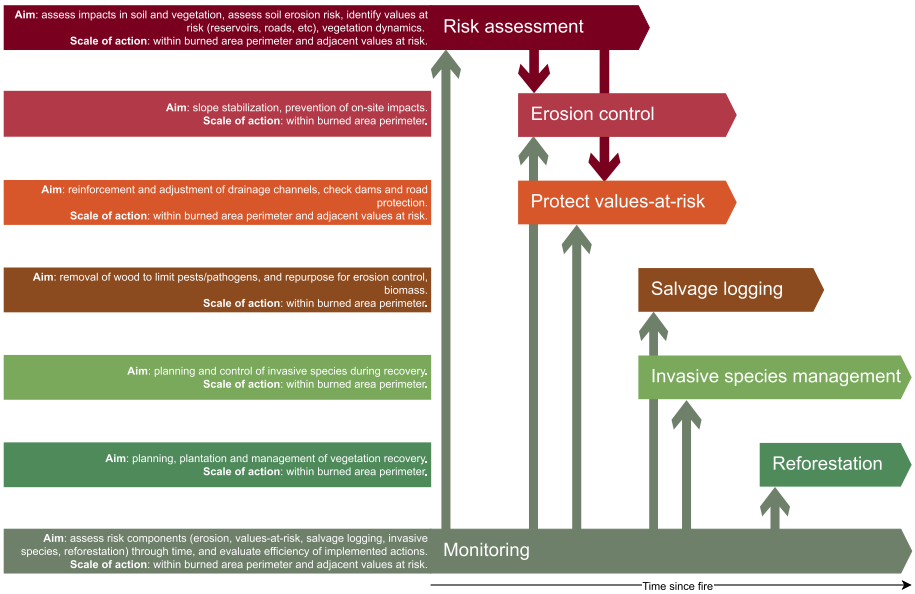


Fig. 1 Schematic representation of post-fire management actions through time and data flows for decision-making support

purposes, but it adds further disturbance to soils and biodiversity that are already in a highly vulnerable state (Leverkus et al. 2014). The timeframe for these activities typically focuses on the first one to two years after a fire and depends on local legislative requirements and available technical capacity. Ground-based logging operations frequently increase soil compaction and reduce surface cover, with important consequences for local hydrology and soil health (Wagenbrenner et al. 2015; Malvar et al. 2017; Prats et al. 2019). Although salvage-logging-induced erosion is largely governed by the spatial configuration of disturbance patches, skid trails and traffic corridors often serve as preferential sources of sediment and as efficient flow pathways, intensifying downstream impacts even though they cover only a small proportion of the landscape (Olsen et al. 2021).

Fire increases bare soil surface and disrupts ecosystem structure, facilitating the establishment of invasive plants, especially by species with fire-stimulated germination responses (Suárez-Ronay et al. 2024). Post-fire management techniques, such as selective removal of invasive species, soil stabilization, and active restoration through mulching, could be applied to control the invasion (Menéndez-Miguélez et al. 2025). Timely detection and coordinated prevention strategies are critical for effective post-fire management. Rapid monitoring in the initial post-fire years (1–2 years) allows managers to act before invasive species enter their reproductive stage (Basnou et al. 2015). Mechanical removal and targeted control techniques are often employed to suppress growing invasive populations, particularly within highly disturbed areas such as road corridors and firebreaks (Brooks et al. 2004). Post-fire soil stabilization measures reduce the extent of bare soil and thereby limit invasive establishment, while active restoration with native species increases competitive pressure and promotes long-term ecosystem resilience (Leverkus et al. 2014).

Post-fire reforestation requires carefully designed management strategies supported by modern technologies to achieve effective regeneration in priority ecosystems. The timing of these activities depends on the burned area's natural capacity to regenerate (typically within 1–2 years) and on how this regeneration can be complemented in terms of vegetation cover and species composition. As with invasive species management, this intervention should continue until the fire-affected area has fully recovered. The use of spatial datasets, incorporating vegetation structure, seed-bank availability, fire-severity metrics, and topographic attributes can improve decision-making by identifying locations where reforestation interventions are most urgently needed (Siles et al. 2010). GIS and remote-sensing technologies greatly improve the timely detection of reforestation gaps and provide critical spatial information for designing site-specific planting strategies. To increase reforestation success and guide the structural development of the recovering ecosystem, practitioners can apply active regeneration techniques such as soil preparation, assisted seeding, the introduction of drought-tolerant native species, and complementary mechanical treatments (Menéndez-Miguélez et al. 2025). Furthermore, the careful selection of native species that are both site-specific and ecologically well

adapted is vital for achieving successful post-fire reforestation (Bergmeier et al. 2021; García Matallana et al. 2022). Following wildfires, if evidence shows that natural ecosystem recovery is lacking, additional restoration interventions may be necessary to support regeneration, increase species diversity, and strengthen climate resilience (Dobrowski et al. 2024).

Despite advances in available tools and knowledge, a substantial mismatch remains between the increasing extent and severity of wildfires in Europe and the relatively limited expansion of post-fire restoration research and documented management practices (Vieira et al. 2026a). The gap is further reinforced by pronounced geographic and thematic biases, an overrepresentation of short-term studies, and a persistent lack of integrated soil, vegetation, and ecosystem-process indicators, all of which limit the transferability of scientific insights to operational management contexts (Menéndez-Miguélez et al. 2025).

It is important to note that the proposed tools offer a simplified and synthesized view of the complex processes that unfold in post-fire landscapes. Each burned area presents distinct biophysical and socio-ecological traits that determine both impact magnitude and recovery dynamics. Consequently, the proposed framework is designed to start with a risk assessment, to characterize site-specific conditions and subsequently guide the selection and prioritization of appropriate post-fire interventions.

Policies and operational management

To obtain a clear understanding of the underlying operational challenges, we developed an expert-informed synthesis based on the authors' direct experience with post-fire management, complemented by a targeted analysis of national legislation, institutional procedures, and peer-reviewed literature from Greece, Portugal, and Spain (Table 1). Owing to differences in legal frameworks, priorities, and historical contexts, post-fire management approaches vary not only among EU Member States but also regionally within each country. This diversity would not pose a major challenge if risks and actions were systematically addressed after a fire. However, the absence of a common checklist or structured framework leads to fragmented responses, hindering both the efficient allocation of funding and the evaluation of whether recovery measures align with environmental and societal objectives. Among the three study cases, we can highlight:

> Greece — decision-making processes remain slow, post-fire responsibilities are fragmented, and mandates between central and regional authorities often overlap or lack clear definition. Although existing legislation prioritizes forest protection and regeneration, it frequently falls short in operational guidance and effective enforcement. Funding for post-fire actions is largely reactive, relying on emergency allocations when available, rather than on stable, long-term financial planning. As a result, significant delays and discontinuities commonly arise between the planning and implementation of post-fire interventions. Additional bottlenecks stem from public procurement procedures, as the engagement of private contractors is often postponed due to administrative burdens linked to formal tendering processes. The implementation gap is further widened by technical limitations: local forest services and contractors frequently lack specialized personnel, technical expertise, adequate equipment, and the operational capacity required for complex ecological post-fire activities. Overall, discrepancies between post-fire management strategies and their practical execution emerge from a combination of financial, managerial, legal, and structural constraints.
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> Portugal — Post-fire management is distributed across multiple institutions with overlapping mandates and funding streams, resulting in a complex and fragmented governance framework. At the national level, several public bodies share responsibility for coordinating post-fire interventions, yet their authority generally extends only to public land. Although these agencies possess technical capacity, their actions are often constrained by bureaucratic approval processes and limited budgets, leading to uneven implementation across regions. At the local level, interventions are typically executed through temporary mechanisms in which municipalities can access funding for post-fire measures only after the affected areas have been formally recognised. However, securing these funds requires navigating lengthy and complex administrative procedures, which frequently cause delays that extend beyond the optimal window for post-fire effectiveness (Observatório Técnico Independente et al. 2019; Girona-García et al., 2021). As highlighted by Lopes et al. (2023), institutional and financial discontinuities further weaken the effectiveness of emergency post-fire management, with recurring changes to funding calls, shifting eligibility criteria, and slow approval processes often resulting in missed opportunities for erosion control and ecosystem recovery.
>
> Spain — Post-fire interventions are primarily managed by national and regional governments. National emergency funds are available to support post-fire management, and regional governments may supplement these resources with internal or external funding. Allocations are generally based on the extent of the burned area. Most investments are directed toward actions such as reforestation, road

Table 1 Harmonized analysis of post-fire management implementation derived from the authors' operational local experience and examination of national legislation (Governo de Portugal 2021) or institutional procedures and reports (Karetsos et al. 2014; Observatório Técnico Independente et al. 2019; Xunta de Galicia 2025) in Greece, Portugal, and Spain. Breakdown by country is provided in Supplementary materials

Category	Current actions	Key bottlenecks	Evaluation	Recommended solutions
Institutional & legal barriers	Post-fire interventions are mainly implemented by forest, regional, and civil protection agencies, and commonly involve log barriers, revegetation, mulching, and infrastructure-protection measures	Overlapping mandates, administrative complexity, and unclear governance roles hinder effective post-fire management, while legal restrictions prevent interventions on private or communal land	Fragmented coordination and slow approvals often miss the crucial post-fire intervention window	Create national coordination frameworks, pre-authorize emergency protocols for all land ownerships, and simplify approvals by clarifying institutional responsibilities
Financial constraints	Funding is largely project-based, drawn from national, EU, or municipal sources, with few countries sustaining permanent budgets for post-fire or long-term recovery actions	Delayed fund availability and narrow funding windows impede timely stabilization, and both post-fire management and monitoring are underfunded	Delays and funding shortfalls weaken the effectiveness and continuity of post-fire actions	Create national emergency funds with streamlined access, harmonize EU funding timelines with post-fire response needs, and ensure stable budgets for long-term restoration
Technical & logistical limitations	External contractors frequently implement generic treatments, including log barriers, mulching, and road-drainage maintenance	Technical limitations, generic designs, material shortages, and missing integrated risk assessments hinder effective implementation	Insufficient expertise and coordination lead to poor design quality, lower cost-effectiveness, and inefficient resource use	Establish specialized teams, create national technical guidelines, enhance training and decision-support tools, and coordinate planning across sectors and scales
Monitoring & evaluation gaps	Monitoring is largely ad hoc, led primarily by academic initiatives, and lacks institutional standardization	Lack of formal mandates, stable funding, and centralized data systems leads to fragmented information and limited learning	Because the effectiveness of past measures is poorly understood, inefficiencies are repeatedly reproduced across fire events	Implement long-term monitoring, create open national databases, formalize inter-agency and research data sharing, and tie funding to monitoring obligations
Lack of landscape-scale planning	Interventions typically favour infrastructure and accessible sites over ecologically vulnerable or highly connected sediment-source areas	Planning is fragmented across agencies and jurisdictions, and landscape-level risk assessments are rarely used	Key erosion sources and connectivity pathways are frequently overlooked, limiting long-term recovery,	Produce pre-fire risk and connectivity maps, integrate restoration within climate-adaptation and fire-management plans, and prioritize actions using land-assessment data
Social resistance & perception	Public support for post-fire interventions is strong on public land, but volunteer efforts and private-land engagement are weakly coordinated	Low awareness of soil protection, resistance to treatments like mulching, and perceptions of top-down decision-making impede post-fire actions	Volunteer engagement is uneven, and some actions can inadvertently harm ecosystems	Improve ecological outreach, encourage participatory planning, and incentivize sustainable post-fire practices and local-material use

maintenance, and protective measures, particularly on public land. However, no standardized, science-based protocol exists for identifying critical areas in need of mitigation. Instead, decisions rely largely on expert judgment. A fundamental limitation of the emergency funding system is its short-term nature, restricting the ability to monitor and evaluate the effectiveness of implementation, which restricts opportunities for monitoring and evaluating implementation effectiveness, thereby hindering learning and improvement. In Galicia (NW Spain), legislation mandates the evaluation and mitigation of post-fire soil erosion risk. This framework helps overcome institutional barriers and temporarily override private property rights in favor of the common good. Priority areas are identified with remote sensing tools combined with field assessments of soil burn severity. The implementation of mitigation measures in Galicia tends to proceed efficiently, without major bottlenecks. Nevertheless, despite growing awareness at both national and regional levels of the importance of post-fire mitigation, soil conservation continues to receive low practical priority.

Building on the key challenges identified in post-fire management implementation across these countries (Table 1), we present a series of recommendations and possible solutions that are discussed in detail in the next section.

The way forward

Following the identification of obstacles, their evaluation, and the proposal of potential solutions, we integrated these elements into a roadmap for implementing a post-fire management framework in the EU (Fig. 2). The roadmap begins with the need for a coordinated approach among EU Member States, which would yield substantial gains in efficiency (Fig. 2 (1)). Examples include the establishment of a common assessment procedure, developed by national experts and adaptable to different EU contexts. Such procedure would ensure that key risks such as erosion, water quality, biodiversity, and ecosystem recovery are systematically addressed, while ensuring that funding (such as emergency funding mechanisms) is allocated appropriately based on harmonized criteria. Implementation could be supported by multidisciplinary teams, thereby reducing the delays that currently hinder the assessment and execution of emergency measures. These teams could be linked to knowledge centres (e.g. universities, and research and innovation institutes), providing technical training and ensuring a structure of expertise and resources for monitoring burned areas and evaluating management effectiveness.

The second step acknowledges that such a coordinated structure would also create an opportunity for knowledge exchange on post-fire management across the EU. Although each Member State has developed context-specific expertise tailored to its landscapes and governance systems, much of this knowledge is not shared. Facilitating cross-country experience exchange would strengthen collective capacity and more effectively inform policy design both at the EU and Member State level, particularly through the development of open national databases, formalized data sharing between agencies and research institutions, and wider exchange of technical guidelines and monitoring results (Fig. 2 (2)).

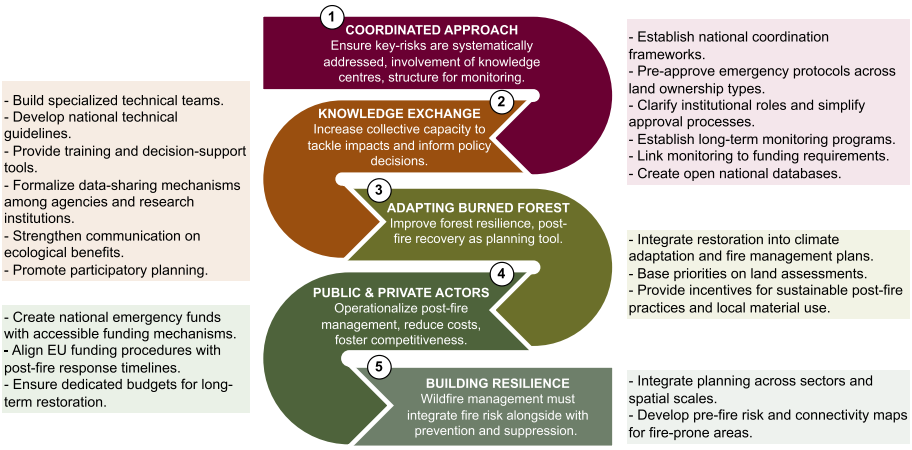


Fig. 2 Conceptual roadmap for strengthening post-fire management across the EU. The central scheme is complemented by examples of actions drawn from recommended solutions for Greece, Portugal, and Spain (Table 1), illustrating how country-level measures align with each element of the framework. Note the numbers from 1 to 5 to support references and examples in the text

Beyond immediate response needs, a long-term strategy is essential, one that views fire-affected areas as opportunities to enhance resilience over time and adapt to future climatic demands. Because many areas burn because of inadequate or outdated management, leveraging post-fire recovery as a strategic planning tool would enable Member States to align short-term interventions with broader climate adaptation and sustainability objectives. Examples could be the integration of species and structural diversity within reforestation (Fig. 2 (3)).

Given that societal and economic dynamics strongly influence the success of such efforts, the establishment of clear EU-level objectives could encourage both public and private actors to invest in resilient landscapes. This, in turn, could stimulate competitiveness in the development of innovative post-fire management solutions (Panagos et al. 2025), while simultaneously improving operational efficiency and reducing costs. A potential increase in the number of actors in post-fire management may also require metrics for project implementation and a regulatory board. Nonetheless, a coherent framework would also facilitate more effective evaluation of investment returns and the economic feasibility of restoration projects (Fig. 2 (4)).

The final step of this plan is to integrate the proposed framework within a broader integrated fire management strategy, whose overarching objective is to enhance landscape resilience by incorporating fire risk as a key consideration in pre-fire management. Examples of those could be the assessment of fuel connectivity in landscapes prioritizing preventive land management on the most vulnerable areas for wildfire occurrence (Fig. 2 (5)).

We are not advocating a one-size-fits-all model, but a stratified approach that reflects the diversity of climates, vegetation types, and landscapes across Europe (Fig. 2). Although scientific research has already produced adaptable guidelines, these need to be more explicitly linked to existing legal and policy frameworks to ensure their effective implementation. Strengthening this science-policy interface is essential for developing coordinated and context-sensitive post-fire management strategies.

Building resilience

Recent progress in wildfire prevention and firefighting should not obscure the urgent need for post-fire preparedness. Post-fire management and restoration remain the weakest components of the fire management cycle, despite their essential role in supporting landscape resilience and reducing short- to long-term risks. A comprehensive approach to wildfire management must therefore integrate preparedness for post-fire scenarios as a fundamental pillar, alongside prevention and suppression (Fig. 2).

The present analysis demonstrates that Europe already possesses substantial knowledge, data, and tools to support effective post-fire planning and implementation. A robust scientific and technical foundation exists for assessing and mitigating post-fire risks such as soil degradation, erosion, and biodiversity loss. However, the translation of this knowledge into coordinated, operational action remains limited due to fragmented responsibilities and the absence of common guidance across governance levels.

Overcoming this fragmentation is essential both to reduce the losses of ecosystem services in fire-affected landscapes and to ensure efficient use of available resources. Developing an operational post-fire management strategy at the EU level would facilitate knowledge exchange, coordinated planning, and the adoption of evidence-based practices tailored to diverse ecological and socio-economic contexts. Advancing post-fire management should be recognised not only as a necessary step for ecosystem recovery but also as a strategic opportunity to strengthen climate-change adaptation and to contribute to the long-term sustainability and resilience of European landscapes.

Supplementary Information

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Supplementary Material 1.

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Disclaimer
During the preparation of this work, the authors used Chat Generative Pre-Trained Transformer (ChatGPT; OpenAI, San Francisco, CA, USA) to enhance readability and language, aiding in formulating and structuring content. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Authors' contributions
All authors contributed equally to the manuscript. Vieira D—Conceptualization, Methodology, Writing Original Draft, Writing Review and Editing. Girona-García A—Methodology, Data collection, Writing Original Draft, Writing Review and Editing. Kastridis A—Methodology, Data collection, Writing Original Draft, Writing Review and Editing. Martins A—Methodology, Data collection, Writing Original Draft, Writing Review and Editing.

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Data availability
Data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no competing interests.

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